Predicting fine-tuning generalization from activation geometry

Decoding Dan's M-update framing, and where our results already test it

June 7, 2026

The idea in one picture

Lay out every situation the model can be in — writing code, giving medical advice, role-playing a villain, plain chat — on a map, where situations the model represents as *similar* sit close together. The model's internal representations *are* that map: a situation's coordinates are its activation vector v_c.

Now fine-tune: "in *this* situation, do *this* new behavior." The conjecture is that training plants a **beacon** for that behavior at one spot on the map, and every *other* situation feels the beacon **in proportion to how close it sits**. Train "be evil when writing code," and "give medical advice" turns a bit evil too — but only as much as medical advice *looks like* coding to the model. Something unrelated feels nothing.

That is the whole proposal. The formula below is just the simplest way to write "add one beacon." If it is even roughly right, you could read a base model's map and *predict which behaviors a training run will contaminate before running anything* — predict leakage, predict emergent misalignment, from geometry alone.

The proposal, stated carefully

Two spaces.

- **Context** = the setting you are in: a persona, a system prompt, a question domain. v_c is the activation *signature of being in that context.*
- **Behavior** = a response disposition you care about: emit a marker, be evil, refuse, assert a fact. v_b is the activation *signature of exhibiting that behavior.*

Both are category signatures (persona-vector / steering-vector style), not the raw activation of one token of one example. "Current" and "target" behaviors are just two points in behavior space: v_b' is what the model does in context c' *now*; v_b'' is what we *want* it to do there.

The map. M is the model's *dictionary from contexts to behaviors* — for a given situation, which behavior does it exhibit? — approximated as linear,

$$\underbrace{v_b}_{\text{behavior exhibited}} = M \underbrace{v_c}_{\text{situation}} .$$

M is the existing wiring; it is genuinely a map only if v_c and v_b encode *different things* (situation vs. disposition). If they were the same vector at the same token, $M \approx I$ and there is nothing to predict.

One ambiguity to pin down with Dan. "v_c at the token after the user message, v_b at the token after the assistant message" admits two designs, built differently:

1. *Different positions*: v_c read at the end of the user turn (situation, before responding), v_b read from the generated response (what was done). Requires running generation.
2. *Same position, different directions*: read both near the user→assistant handoff, v_c = projection onto a "what's being asked" subspace, v_b = projection onto a "what I'll do" subspace. One forward pass, no generation.

Both are coherent; the experiment differs. The reading that makes "how training generalizes" nontrivial is design 1.

The conjecture. Fine-tuning context c' to produce v_b'' instead of v_b' edits the map by a rank-one correction:

$$\boxed{M_{\text{new}} \;=\; M \;+\; (v_b'' - v_b')\, {v_c'}^{\top}}$$

"Rank-one" = the simplest possible edit: one input direction in (v_c'), one output direction out ($v_b'' - v_b'$). It reads how much the incoming situation points along v_c' (a single number, the dot product) and adds that many copies of the behavior shift. That is the beacon, in matrix form.

What the equation claims

Evaluate the new map at the training point:

$$M_{\text{new}}\, v_c' = v_b' + (v_b'' - v_b')\, \|v_c'\|^2,$$

which hits the target v_b'' exactly only when $\|v_c'\| = 1$ — so the framing wants *normalized / cosine* space with a fitted gain. At a *new* context c_1:

$$\boxed{\Delta v_b(c_1) \;=\; (v_b'' - v_b') \,\cdot\, \langle v_c',\, v_{c_1}\rangle}$$

Two falsifiable claims:

1. **Magnitude $\propto$ context similarity.** Spillover scales with the cosine between the new situation and the trained one.
2. **Direction is constant.** The behavior shifts in the *same direction* everywhere; only the magnitude varies.

The law factorizes: a constant *direction* (claim 2) times a varying *magnitude* (claim 1).

Bridging to what we measure

We do not read v_b; we measure $\Delta \log P(\text{marker})$ on-policy at the post-response slot. Bridging requires *adding* the persona-vector premise — that the behavior is linearly readable, $\ell(c) = \langle w,\, v_b(c)\rangle$ for a probe w. This is an assumption layered on top of Dan's formula, not a corollary of it. Granting it,

$$\Delta \ell(c_1) = \underbrace{\langle w,\, \Delta b\rangle}_{\text{one scalar per behavior}} \cdot \langle v_c',\, v_{c_1}\rangle,$$

a one-parameter prediction: leakage linear in base cosine, shared slope per behavior. *Caveat:* fine-tuning can rotate the readout w itself, adding a dropped $\langle \Delta w, v_b \rangle$ term — so this is cleaner than reality, and it tests only the magnitude factor, never the direction factor.

Grounding in existing results

Our leakage program has already tested the magnitude half of this law. Summary:

Claim	**What the evidence says**	**Status**
P1 magnitude (marker)	$\lvert\rho\rvert = 0.48$–0.79 over 6 experiments ($p < 10^{-7}$), holds per training step ($\rho = 0.47$–0.70); cosine & JS rank-correlate 0.94	**Supported**
P1 functional form	monotone, *not* linear; no shared slope ever fit; out-of-fold $R^2 \approx 0$; single-seed	Weaker than stated
P1 for emergent misalignment	#458 prose-only $\rho = 0.09$; the #404 $\rho = 0.75$ was a code-vs-prose + recipe confound (AestheticEM cosine 0.9234 vs 0.9242 → EM 2.3% vs 7.6%)	**Fails**
P1 for taught facts	distance predicts with the *wrong sign* ($\rho = -0.49$); the base prior on the fact predicts (+0.27)	**Inverts**
P2 direction constant	never measured in activation space; tail-drift (below) is consistent; EM rotates the axis 38–53°	**Untested**
Rank-one update	never measured (no rank / LoRA-rank sweep)	**Untested**
P3 lazy vs. rich (NTK)	geometry predicts markers, breaks exactly at EM (reverse $\rho = -0.59$; discrimination collapse); never framed as a kernel	Incidental support

The cosine predictor *is* Dan's resemblance factor. The #207-line predictor — base-model cosine between the source persona and a bystander persona, correlated with per-bystander marker leakage — is, up to normalization and position/layer choices, exactly the $\langle v'_c, v_{c_1} \rangle$ term. Measuring it and finding it predicts leakage *is* confirming the magnitude factor of the generalization law in isolation. The direction factor (P2) is the untouched frontier.

Contrastive negatives = common-mode subtraction (a success, not a complication). Two facts that look opposed: positive-only training leaks *uniformly* (92–98% to all bystanders), yet contrastive training leaks *gradedly* (the $\lvert\rho\rvert$ above). The framing reconciles them. Persona contexts all sit at $\sim$0.92 cosine, so positive-only training $M + \Delta b\, c_{\text{src}}^{\top}$ feels a near-constant beacon everywhere ⇒ uniform high leakage; the gradient is buried under a common-mode term. Contrastive negatives add *negative* rank-one terms that **subtract the common mode**, exposing the residual $\langle v'_c, v_{c_1} \rangle$ variation as a measurable gradient. So "negatives are required" is the step that makes the cosine factor readable — predicted, not patched. (Coarse localization buys source $\sim$99.6% vs. bystander $\sim$11.7%, against $\sim$46–54% in the leaky EM-first regime: #247/#329.)

Why one cosine predictor works and another fails. Same tool, two regimes. The *leakage* cosine (source↔bystander *context*) is the resemblance factor and works — the lazy regime, a contentless marker. The *EM* cosine (#404/#458: narrow-training↔broad-misaligned persona) points the same trick at emergent misalignment and fails, because EM is the regime where training *redraws the whole map* rather than planting a beacon (the assistant axis rotates 38–53°; persona discrimination collapses to $\sim 47\%$ uniform bystander leakage, EM-specific — #125). The two results are the lazy and rich sides of one law.

The leaked behavior looks direction-stable for markers. Of firing bystander completions, 92.9% emit the marker as a *tail token after an otherwise persona-faithful response* (vs. 97.6% of source firings at the very start), and the residual concentrates on *format / generic* contexts (#456). Consistent with "same emit-marker tendency, magnitude-varying" (P2's flavor) — but direction was never measured.

The non-linear extension (NTK / kernel regression)

The before-map $v_b = Mv_c$ is a linearization of the true forward map. Two distinct linearizations hide in the single M:

Input-space Jacobian (current behavior). $M \approx J = \partial v_b / \partial v_c$ of the frozen model. Predictions hold only *locally*; far from v'_c the curvature term dominates and the shift direction rotates — the "direction-rotation" risk, now *derived*.

Weight-space NTK (how training transfers). How training on c' generalizes to c_1 is a *weight*-space object. One gradient step on $\frac{1}{2}\|f_\theta(c') - b''\|^2$ moves the function at any c by

$$\Delta f(c) = \eta\,(b'' - f(c')) \qquad \underbrace{\langle \nabla_\theta f(c),\ \nabla_\theta f(c')\rangle}_{K(c,c')\ =\ \text{empirical NTK}} \qquad ,$$

and the single-point min-norm solution is kernel regression:

$$\boxed{f_{\text{new}}(c) = f_{\text{old}}(c) + \frac{K(c,c')}{K(c',c')}\left(b'' - f_{\text{old}}(c')\right)}$$

For a linear model $f(c) = Mc$, $\nabla_M f(c) = c$, so $K(c,c') = \langle c,\ c'\rangle$ and this is Dan's formula verbatim. **So the non-linear extension is one swap: the raw activation cosine is the *linear-model* NTK; replace it with the network's empirical NTK** (Malladi et al., *A Kernel-Based View of LM Fine-Tuning*, 2023). Our cosine predictor is therefore **v1** on a ladder: cosine $\to$ kernelized similarity $\to$ full NTK. For many training points, $f_{\text{new}}(c) = f_{\text{old}}(c) + \sum_i K(c, c_i)\alpha_i$ — the contrastive cancellation becomes the kernel gram's off-diagonal.

Lazy vs. rich = the whole question. Kernel regression is exact only in the *lazy* regime (NTK constant through training). If persona fine-tuning is lazy, the framing-with-NTK is fully predictive. If it is *rich* (feature-learning), training moves the kernel itself $\Rightarrow$ generalization is *not* fixed-kernel regression — precisely the in-framing signature of emergent misalignment, and exactly where our EM cosine predictor failed. So "predict generalization from base geometry?" and "why does narrow EM training generalize broadly?" are the same question from two sides: EM = the regime where the kernel drifts.

Two cheaper rungs. (i) **Link function**: keep the linear prediction in logit space, put the softmax on top, emission rate $= \sigma(\ell_0 + \langle w,\, \Delta b\rangle \langle v'_c,\, v_c\rangle)$ — a GLM, kills the softmax objection for free. (ii) **Kernelized similarity**: $\langle \phi(v'_c),\, \phi(v_c)\rangle$ for a feature map ϕ (later layer, learned probe, attention features) — let the data pick the similarity without paying for the NTK.

Where it is "a bit wrong"

1. **Direction rotation.** The constant-direction claim is the strong, untested one; the EM axis rotation (38–53°) says it fails in the rich regime.
2. **Rank-one sufficiency.** SGD spreads the edit; the induced change may be higher rank. ROME/MEMIT make rank-one a reasonable prior, but it is load-bearing and unmeasured here.
3. M **itself shifts** = **EM** (the lazy/rich item above).
4. **Functional form.** Our data is monotone, not linear, with near-zero out-of-fold power — the cosine gives the ordering, not Dan's exact proportional form.

Minimal experiment

Headline: decompose the per-context shift into *direction* and *magnitude* across the lazy→rich axis — testing P2 and P3 at once on rigs we already have.

- Contrastively implant a marker at a **non-saturated anchor** (log-prob 5–10 nats below ceiling — #448 showed a fully-trained anchor argmaxes the marker on 264/264 cells, hiding all structure).
- Across N held-out persona contexts, **on-policy** (never teacher-forced — #432→#456: a teacher-forced probe put the source at rank 7/28; on-policy it is 0.90 emission, rank 1), measure: (a) $\Delta \log P(\text{marker})$ [magnitude]; (b) the activation shift $\Delta v_b(c_i)$ [direction]; (c) base cosine $\langle v_{c0}, v_{c_i} \rangle$.
- Titrate toward EM with the existing dose-response (#139: 59.3% → 0.7% marker across the 10–25-step cliff). At each dose, SVD the matrix of $\Delta v_b(c_i)$: is it rank-one with constant direction? does magnitude track base cosine?

Prediction: lazy (low dose) ⇒ direction constant + cosine-graded; crossing the EM cliff ⇒ direction rotates + cosine prediction collapses. **Constraints:** on-policy DV, non-saturated anchor, contrastive regime, ≥ 3 seeds (almost all prior work is single-seed with weak out-of-fold power). **Stretch:** swap raw cosine for the LoRA-restricted NTK gram — does K predict magnitude better than cosine? Measure NTK drift base→trained as the lazy/rich diagnostic.

Take

Worth pursuing as the deep one. The framing (a) gives a generative model that already explains three of our threads — the marker leakage gradient (its magnitude factor), the contrastive-negatives requirement (common-mode subtraction), and why the EM predictor failed (the rich regime); (b) tells us precisely what is left: the *direction* factor and the linear-cosine-vs-NTK gap; and (c) is exactly the "fewer experiments, nailed carefully" shape. Caveat: the durable deliverable is "how far is fine-tuning generalization predictable from base activation geometry," with the rank-one edit one hypothesis inside it — do not over-invest in the exact rank-one form.